Big Data Analytics
MBA Group Project - NMIMS (20 Marks)

Project Objective

You have been appointed as a Data Analytics Consulting Team for a rapidly growing organization experiencing challenges due to increasing data volume, velocity, and complexity.

Your task is to design, build, and demonstrate an end-to-end Big Data Analytics solution that transforms raw data into actionable business decisions.

Rather than focusing only on algorithms, the project should demonstrate how modern data engineering and analytics platforms enable organizations to make faster, smarter, and more scalable decisions.

By the end of the project, your team should answer three executive questions:
• What business problem are we solving?
• How did we engineer the data pipeline to solve it at scale?
• What decisions should management take based on the insights generated?

Learning Outcomes

Business Analytics (CLO 3)
• Translate an ambiguous business problem into an analytical problem.
• Identify meaningful KPIs.
• Generate actionable recommendations.
• Communicate insights suitable for senior management.

Big Data Engineering (CLO 4)
• Build scalable data pipelines.
• Process large datasets using distributed computing.
• Engineer features suitable for machine learning.
• Apply modern cloud-based analytics environments.

Business Scenario

Choose one real-world business problem where spreadsheet-based analysis is no longer practical. Suitable domains include Ride Sharing, E-Commerce, Banking, Healthcare, Insurance, Retail, Manufacturing, Telecommunications, Logistics, Smart Cities, Education, Sports Analytics and Social Media. Use a structured, semi-structured or unstructured dataset representative of a real business environment.

Expected Solution Architecture

Business Problem → Data Sources → Data Ingestion → Distributed Storage → Data Engineering → Feature Engineering → Machine Learning → Business Insights → Executive Recommendations

Phase 1 – Enterprise Data Acquisition & Storage (Sessions 1–9)

Design a scalable data ingestion and storage strategy using technologies such as HDFS, Sqoop, Flume, cloud storage, APIs or equivalent mechanisms. HBase and Kafka are optional where appropriate. Explain data sources, storage architecture, movement of data and scalability considerations.

Phase 2 – Data Engineering & Preparation (Sessions 10–13)

Using Databricks and PySpark:
• Load and prepare the dataset.
• Clean, filter and join data.
• Handle missing values and outliers.
• Perform feature engineering.
• Encode categorical variables and apply scaling where appropriate.
Justify each engineering decision.

Phase 3 – Distributed Machine Learning (Sessions 14–18)

Implement at least two analytical approaches using PySpark MLlib.

Regression: Linear Regression, Decision Tree, Random Forest or Gradient Boosting.
Classification: Logistic Regression, Decision Tree, Random Forest, Naïve Bayes, Support Vector Machine or Gradient Boosting.
Clustering: K-Means, Bisecting K-Means or Gaussian Mixture Models.

Compare model performance using appropriate evaluation metrics and justify the final model selection.

Executive Deliverables

1. Executive Presentation (Max 10 slides Including Start & Thank you slide)
• Business problem
• Industry background
• Dataset overview

• Solution architecture
• Key findings
• Model performance
• Business recommendations
• Risks and future roadmap

2. Technical Implementation
• Databricks Notebook, PySpark Notebook or Jupyter Notebook
• Well documented and fully executable

3. Consulting Report (PDF)
• Executive Summary
• Business Context
• Data Understanding (5 Vs)
• Enterprise Architecture
• Data Engineering
• Machine Learning
• Business Insights
• Strategic Recommendations
• Appendix (References, AI usage disclosure, Team contributions)

Evaluation Rubric

Evaluation Criteria	Marks
Business Problem Definition & Executive Storytelling	3
Data Engineering & Architecture	4
Feature Engineering & Data Preparation	3
Machine Learning Implementation	4
Model Evaluation & Interpretation	2
Business Recommendations & Strategic Value	2
Professionalism, Documentation & Presentation	2

Interview Value

Students should present this project as a consulting-style analytics engagement demonstrating their ability to translate business problems into scalable data solutions, engineer enterprise-grade data pipelines, build machine learning models, and communicate executive-level recommendations.

Important Dates & Milestones

• Project Topic Submission: 1 August
• Faculty approval and mentoring window: 1 August – 10 August
• Students should coordinate discussion slots through their Class Representative (CR).
• Professor Office Hours for Project Discussions:
 - Monday: 10:00 AM – 10:30 AM
 - Wednesday: 10:00 AM – 10:30 AM
 - Friday: 10:00 AM – 10:30 AM
• Final Report, Code Repository and Presentation Submission: 8 September (Tentative)
• Project Presentations (Tentative): 11 September or 12 September (final schedule will be communicated separately).

Team Formation & Communication

• Maximum team size: 6 students.
• Students are strongly encouraged to form teams using their existing college project/student groups.
• Avoid requesting individual (1:1) meetings with the professor.
• Each team should nominate one (or at most two) team leads to represent the group during discussions if the entire team cannot attend.
• All project-related communication should be coordinated through the Class Representative (CR).

Repository & Coding Standards

• Every team must create and maintain a GitHub repository from the start of the project.
• The repository should include meaningful commit history demonstrating contribution throughout the trimester.
• Maintain a clear folder structure for notebooks, datasets, documentation, presentations and reports.
• All code should preferably follow Python coding standards (PEP 8), include comments where necessary, and be modular, readable and reproducible.
• Include a comprehensive README with project objectives, setup instructions, architecture and execution steps.

Sample Datasets (Illustrative Only)

Teams may select any suitable large-scale dataset. Example sources include:

• Kaggle (https://www.kaggle.com/datasets)
• Uber Pickups
• Amazon Product Reviews
• Instacart Market Basket Analysis

- Online Retail II
- Credit Card Fraud Detection
- NSE Historical data
- MIMIC Healthcare Dataset (where appropriate)
- MovieLens
- Spotify Tracks
- Airline On-Time Performance
- IPL / Cricket Analytics
- FIFA Player Statistics

Students are encouraged to choose datasets that naturally lend themselves to business insights and machine learning rather than selecting datasets solely for technical complexity.